

Esther Keerthi Suravarapu

Boston, MA 02120 | suravarapu.e@northeastern.edu | [LinkedIn](#) | [GitHub](#)

Available: Jan – Jun 2027

EDUCATION

Northeastern University, Boston, MA (Sept 2025 – Present)

Khoury College of Computer Sciences (Expected Graduation: May 2027)

Candidate for a Master of Science in Artificial Intelligence **GPA: 3.67/4.0**

Related courses: Reinforcement Learning & Sequential Decision Making, Foundations of Artificial Intelligence

GITAM University, Hyderabad, India

Bachelor of Technology in Computer Science, 2025

GPA: 7.92/10

Related courses: Machine Learning, Deep Learning, Data Structures, Algorithms, Databases

TECHNICAL KNOWLEDGE

Languages: Python, SQL, JavaScript, Java, HTML/CSS

Databases: PostgreSQL, MySQL, Redis, ChromaDB (vector DB)

Skills: ML Engineering, Reinforcement Learning, NLP, RAG Systems, MLOps, Deep Learning, LLM Fine-Tuning

Libraries: PyTorch, TensorFlow, XGBoost, LightGBM, HuggingFace Transformers, PEFT, LangChain, LangGraph, SHAP, Optuna, FastAPI, Flask

Tools: Docker, GitHub Actions (CI/CD), MLflow, Evidently AI, Feast, LangSmith, Git, Linux, pytest

WORK EXPERIENCE

Spoors Technologies Pvt. Ltd., Hyderabad, India

AI Intern (May 2024 – Jul 2024)

- Trained an XGBoost classifier to predict overdue field tasks, reaching 84% accuracy (vs. 61% majority-class baseline) and F1 = 0.79 on imbalanced task-completion data for the EFFORT workflow platform
- Built a RAG pipeline (LangChain + vector DB) connecting LLM queries to a proprietary knowledge base via REST API middleware for context-aware field-worker support
- Developed an NLP classification pipeline with HuggingFace Transformers that auto-sorted support tickets into 8 categories, cutting manual triage time ~20%; served via Flask API
- Cleaned ~40,000 user activity logs (missing timestamps, duplicate API events) in Pandas, reducing model training time ~30%

Enoch Industries Pvt. Ltd., Hyderabad, India

Web Developer Intern (Sep 2023 – Feb 2024)

- Shipped a customer-facing dashboard and task-assignment feature in React + Node/Express across a 6-month release cycle, handling ~500 daily active users with real-time status updates

PROJECTS

Healthcare Agentic RAG System | Solo Project (Jun 2026)

- Designed a three-agent pipeline (Retriever → Validator → Synthesizer) via LangGraph for clinical Q&A on PubMedQA, achieving Recall@5 = 1.00; deployed to Hugging Face Spaces
- Built a Validator agent that catches topically-adjacent but non-answering chunks — a failure mode invisible to recall metrics — and triggers automatic re-retrieval before synthesis
- Containerized with Docker (multi-stage build), served via FastAPI, tested at >70% coverage, CI/CD via GitHub Actions

Production MLOps Fraud Detection Pipeline | Solo Project (Jul 2026)

- End-to-end fraud detection on a 577:1 imbalanced dataset; chose PR-AUC as primary metric, tuned XGBoost via Optuna (50 trials) to PR-AUC = 0.8648 with full MLflow experiment tracking
- Integrated SHAP interpretability, Evidently AI drift monitoring (covariate/concept/label), and Feast feature store for offline→online serving; deployed on Render with CI/CD
- Wrote a system-design doc for scaling to 10M transactions/day covering load balancing, streaming feature pipelines, A/B testing, and retraining triggers

Dueling DQN Reproduction | CS5180 Capstone, Team of 2 (Jul – Aug 2026)

- Reproducing Wang et al. (ICML 2016) Dueling Network Architectures: implementing and evaluating the corridor environment against the paper's original results in PyTorch
- Debugging network construction, target-network sync, and replay buffer logic across DQN and Dueling DQN variants

Video Captioning with CNN-LSTM + Attention | Team Lead, 4 members (Sep 2024)

- Built a video captioning system in TensorFlow achieving 85% BLEU (20% over baseline); optimized inference to cut latency 30%